Artistic ability
17-000-year-old art from the Lascaux cave in France shows a high level of sophistication. Modern humans created these images that we can still relate to today.

considerable intellectual reasoning. The discovery of a hyoid bone in Kebara Cave, Israel, dating to about 60,000 years ago, intensified the debate about Neanderthal linguistic abilities. The Kebara hyoid is almost identical to that of modern humans, which has led some anthropologists to claim that the Neanderthals were capable of fully articulate speech. Others disagree, pointing to the high position of the larynx, which would limit the sounds they could make. Some believe that Neanderthals had the communication skills of modern infants. The controversy is unresolved, but most scientists agree that Neanderthals did not have the advanced linguistic and communication skills of *Homo sapiens*.

Blombos beads
These 75,000-year-old perforated shell beads from Blombos Cave, South Africa, are perhaps the oldest known human ornaments in the world.

The great leap

Human language may have evolved because of the need to handle increasingly complex social information, perhaps about 250,000 years ago. As group sizes increased, so did an ability to learn language that could be used to articulate social relationships. It was only later—perhaps around 40,000 years ago during a time that has been referred to as the "Great Leap Forward"—that modern humans developed language of the kind we would recognize today.

Cultural explosion

Connected to the development of speech is the arrival of cognitive thought in early humans. This includes qualities such as perception of our place in the world, intelligence, and moral codes that come with more elaborate societies. None of these advances would have been possible without sophisticated speech. We don't know when *Homo sapiens* acquired the conscious thought and the abilities we have today, but it was at least 40,000 years ago, and most likely in tropical Africa or southwest Asia. It appears that conscious thought evolved after modern human anatomy, for *Homo sapiens* flourished in tropical Africa at least 160,000 years ago, long before the appearance of the elaborate art traditions of the late Ice Age.

First artists

The creation of art requires reasoning and an ability to plan ahead and express intangible feelings. Some of the earliest known decorated artifacts, which were found in Blombos Cave in South Africa, are about 75,000 years old (see left) and are very basic. The full range of human artistic skills came into play during the late Ice Age, epitomized by the cave art, jewelry, sculpture, and carving of the Cro-Magnon people of Western Europe (see pp.26–27). The great bulls at Lascaux cave in France, and the polychrome bison at the cave at Altamira, in Spain, reflect human societies with complex religious beliefs and relationships with the spirit world. Although we do not know exactly what these paintings mean, it is clear that they had great symbolism for those who painted them. This knowledge would have been passed down through the generations by speech and song. For all later human societies, art has remained an important way of expressing our beliefs and knowledge of the world.

AFTER

Sophisticated levels of speech developed as society became more complex. Written records also became important as a method of communication.

POWER THROUGH SPEECH
Speech and language enhanced cooperation between hunters, which led to the greater success of human societies around the world. **Groups could plan** game drives, negotiate exchanges of toolmaking stone, and **share intelligence** about food and water supplies.

EGYPTIAN WRITING

KEEPING RECORDS
Cuneiform writing **62–63 »** developed in West Asia *c.* 3000 BCE as a means of recording commercial transactions and inventories. **Egyptian hieroglyphs** developed at around the same time.

WRITING HISTORY
By the end of the 3rd millennium BCE, writing was widely used for **recording history, philosophy, and science 102–103 »**.

PASSING ON KNOWLEDGE
Speech and writing allowed knowledge and cumulative experience to be passed on from generation to generation.

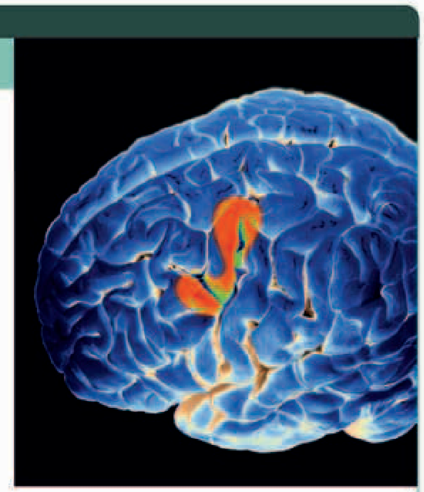
ABSTRACT THINKING
Today, symbols such as **road signs** are part of an **internationally understood language** we use every day.

ROAD SIGN SYMBOL

HOW WE KNOW

BRAIN DEVELOPMENT

Research into the brain can reveal some evidence about the development of speech. Soft brain tissues do not fossilize, and are only preserved in casts of the inside of the skull case. The earliest signs of development of Broca's area, the part of the brain that controls speech, occur in *Homo habilis* about two million years ago. *Homo erectus* also shows signs of development in Broca's area, perhaps an indication of slowly evolving speech. However, any study of language abilities from casts is tentative. Unless a well-preserved hominin brain is discovered—which is unlikely—the amount that we are able to discover from Broca's area is limited, and tangible evidence from hyoid bones will still be needed to learn about fluent speech. Much remains speculative in our knowledge of the evolution of speech.



Understanding speech production
Broca's area—showing up red on this brain scan—is located in the left hemisphere of the frontal lobe. As our knowledge of the human brain grows, so does our understanding of how speech developed.